

Y-P Fibonacci to Espresso F Galois Transformation

1 Purpose of the Code

This program transforms the Y-P Fibonacci LFSR-based stream cipher into an equivalent stream cipher based on the Espresso Galois NFSR F using the modified Y-P Fibonacci-Type II Galois transformation algorithm.

The primary objective of the implementation is to compare the output sequences generated by the original Fibonacci configuration and the transformed Galois configuration, ensuring functional equivalence under different compensation strategies.

Two types of compensation techniques are considered during the transformation process:

- Simple Compensation
- Iterative Compensation (performed in descending order)

The code evaluates and compares the output sequences for four different combinations of these compensation methods to study their correctness and behavioral equivalence.

The combinations examined are:

1. Output function: Iterative, Initial state: Simple
2. Output function: Simple, Initial state: Iterative
3. Output function: Iterative, Initial state: Iterative
4. Output function: Simple, Initial state: Simple

2 Sage Files Corresponding to Different Combinations

In the GitHub repository, there are four Sage files supporting the above combinations.

1. **Output function: Iterative, Initial state: Simple**

Corresponding Sage file:

`Fibo2GaloisF_IterativeOutputFn_SimpleInitState.sage`

2. **Output function: Simple, Initial state: Iterative**

Corresponding Sage file:

`Fibo2GaloisF_SimpleOutputFn_IterativeInitState.sage`

3. **Output function: Simple, Initial state: Simple**

Corresponding Sage file:

`Fibo2GaloisF_SimpleOutputFn_SimpleInitState.sage`

4. **Output function: Iterative, Initial state: Iterative**

Corresponding Sage file:

`Fibo2GaloisF_IterativeOutputFn_IterativeInitState.sage`

3 Input Parameters

To run the program, the following inputs are required.

Table 1: Input Parameters

Inputs	Variable used in the program
No. of bits (256)	<code>n</code>
Feedback function of Yao-Parampalli Fibonacci LFSR	<code>fibo_feedback_fn</code>
Monomials to shift	<code>monomials</code>
Y-P Fibonacci output function (2289 monomials)	<code>fz_fibo</code>
Random initial state (256)	<code>X_initial</code>

4 Output Parameters

The program generates the following outputs.

Table 2: Output Parameters

Outputs	Variable used in the program
Feedback function of Espresso F Galois NFSR	<code>gal_feedback_fn</code>
Combined compensation list	<code>C</code>
Espresso F Galois output function	<code>fz_gal</code>
Compensated initial state	<code>Xhat</code>
Y-P Fibonacci keystreams	<code>outputs_fib</code>
Espresso F Galois keystreams	<code>outputs_gal</code>

Finally, the program compares the Y-P Fibonacci keystreams and the Espresso F Galois keystreams. If they match, the program returns `True`, otherwise `False`

5 User Defined Functions

The following user-defined functions are used in the implementation.

Table 3: User Defined Functions

Purpose	Function used in the program
To shift monomials from $f(a)$ to $f(b)$	<code>shift_term</code>
Compute compensation lists for each monomial separately, and also the combined result	<code>compute_compensation</code>
Iterative compensation for output function	<code>apply_compensation_iterative</code>
Simple compensation for initial state	<code>initialValue_simple_compensate</code>
Simple compensation for output function	<code>apply_compensation_simple</code>
Iterative compensation for initial state	<code>initialValue_iterative_compensate</code>
To collect information of the output function	<code>getInfo</code>
To generate the cipher keystreams	<code>cipher_step</code>

6 Software Environment

All programs run on SageMath version 10.7, Release Date: 2025-08-09.

7 Default Outputs Printed by the Program

After running the program, the following default parameters are printed. If necessary, please check Tables 1 and 2 for other parameters to print.

1. Compensation method used for the output function and the initial state

2. Y-P Fibonacci feedback function
3. Espresso F Galois feedback function
4. Information of Y-P Fibonacci output function
5. Information of Espresso F Galois output function
6. Random initial state (for Y-P Fibonacci)
7. Compensated initial state (for Espresso F Galois)
8. Output sequence (Y-P Fibonacci)
9. Output sequence (Espresso F Galois)
10. Matching status of Y-P Fibonacci keystreams and Espresso F Galois keystreams